Noncommutative Davis inequalities and applications

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Davis inequalities

- Let $(\Omega, \mathcal{F}, \mathbb{P}; \{\mathcal{F}_n\}_{n\geq 1})$ be a probability space associated with an increasing filtration $\{\mathcal{F}_n\}_{n\geq 1}$.
- Let $\{\mathbb{E}_k\}_{k\geq 1}$ be the conditional expectation with respect to $\{\mathcal{F}_n\}_{n\geq 1}$.
- For a scalar martingale $f_n = \mathbb{E}_n f$, let $d_n f = \mathbb{E}_n(f) - \mathbb{E}_{n-1}(f)$, and

$$f^* = \sup_{n\geq 1} |f_n|, \qquad S(f) = \left(\sum_{n\geq 1} |d_n f|^2 \right)^{1/2}.$$

The Burkholder–Gundy inequalities compare maximal and square functions for $1 < p < \infty$.

Theorem (Burkholder, 1966)

There exist $c_p, C_p > 0$ such that

$$c_p \|S(f)\|_p \leq \|f^*\|_p \leq C_p \|S(f)\|_p, \quad 1 < p < \infty.$$

Davis inequalities

Definition

A filtration is regular if for each n

$$\mathbb{E}_n(f) \leq c_{\mathrm{reg}}\, \mathbb{E}_{n-1}(f), \qquad f \geq 0,$$

with one constant c_{reg} independent of a and n.

Theorem (Burkholder–Gundy, 1970, Acta Math.)

Under the regular condition, there exists $c_p, C_p > 0$ such that

$$c_p \left\|S(f)\right\|_p \leq \left\|f^*\right\|_p \leq C_p \left\|S(f)\right\|_p, \quad 0 < p \leq 1.$$

Theorem (Davis, 1970)

For general martingale, there is a universal constant $C > 0$ such that

$$C^{-1} \left\|S(f)\right\|_1 \leq \left\|f^*\right\|_1 \leq C \left\|S(f)\right\|_1.$$

Here we study the Davis inequality for noncommutative martingales

Noncommutative martingale theory

- I. Cuculescu (1971). *Martingales on von Neumann algebras*. J. Multivariate Anal. **1**(1), 17–27.
- G. Pisier and Q. Xu (1997). *Non-commutative martingale inequalities*. Comm. Math. Phys. **189**(3), 667–698.
- M. Junge (2002). *Doob's inequality for non-commutative martingales*. J. Reine Angew. Math. **549**, 149–190.
- N. Randrianantoanina (2002). *Non-commutative martingale transforms*. J. Funct. Anal. **194**(1), 181–212.
- D. Potapov, F. Sukochev and D. Zhou (2024). *Noncommutative nest Hardy spaces and martingales*. Int. Math. Res. Not. IMRN **2024**(21), 13724–13757.
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Noncommutative martingales

• Let $(\mathcal{M},\tau)$ be a finite von Neumann algebra with a normal faithful normalized trace. For $0<p<\infty$,

$$\|x\|_p=\tau(|x|^p)^{1/p}.$$

• Consider an increasing, weak*-dense filtration

$$\mathcal{M}_1\subset\mathcal{M}_2\subset\cdots\subset\mathcal{M},\qquad \mathcal{E}_n:\mathcal{M}\longrightarrow\mathcal{M}_n,$$

where each $\mathcal{E}_n$ preserves the trace.

• For convenience, we here only consider the noncommutative martingale $x:=(\mathcal{E}_n(x_{n+1}))_n=(x_n)_{n\geq 1}$. Its differences are

$$d_1x=x_1,\qquad d_nx=x_n-x_{n-1}\quad(n\geq 2).$$

Nc Burkholder-Gundy

Noncommutativity produces two square functions:

$$S_c(x) = \left(\sum_{n \geq 1} |d_n x|^2 \right)^{1/2}, \qquad S_r(x) = \left(\sum_{n \geq 1} |d_n x^*|^2 \right)^{1/2}.$$

Let

$$\|x\|_{H_p^c} = \|S_c(x)\|_p, \qquad \|x\|_{H_p^r} = \|S_r(x)\|_p.$$

Theorem (Pisier and Xu, 1997)

For $1 < p \leq 2$,

$$L_p(\mathcal{M}) \sim H_p(\mathcal{M}) := H_p^c(\mathcal{M}) + H_p^r(\mathcal{M}),$$

$$\|x\|_{H_p} = \inf_{x=y+z} \left\{ \|y\|_{H_p^c} + \|z\|_{H_p^r} \right\},$$

where y and z are martingales.
For $2 \leq p < \infty$,

$$L_p(\mathcal{M}) \sim H_p(\mathcal{M}) = H_p^c(\mathcal{M}) \cap H_p^r(\mathcal{M}),$$

$$\|x\|_{H_p} = \max \left\{ \|y\|_{H_p^c}, \|z\|_{H_p^r} \right\},$$

Nc Doob inequality

The noncommutative maximal norm uses factorizations in place of a pointwise supremum:

$$x_n = a\, v_n\, b, \qquad a, b \in L_{2p}(\mathcal{M}), \qquad (v_n) \subset \mathcal{M}.$$

$$\|(x_n)\|_{L_p(\mathcal{M},\ell_\infty)} = \inf_{x_n = a v_n b} \left\{ \|a\|_{2p} \sup_n \|v_n\|_\infty \|b\|_{2p} \right\}.$$

Positive sequences

If every $x_n \geq 0$, then

$$\|(x_n)\|_{L_p(\mathcal{M},\ell_\infty)} = \inf \left\{ \|A\|_p : A \in L_p(\mathcal{M})_+,\ x_n \leq A \text{ for all } n \right\}.$$

Thus a positive majorant A replaces the classical maximal function.

Nc Doob inequality

Junge's noncommutative Doob inequality gives

Theorem (Junge 2002)

$$\|(\mathcal{E}_n x)_n\|_{L_p(\mathcal{M},\ell_\infty)} \leq C_p \|x\|_p, \qquad 1 < p \leq \infty.$$

Junge and Xu identified the optimal growth $C_p \asymp (p-1)^{-2}$ as $p \downarrow 1$.

Failure of one direction at $p = 1$

Junge and Xu (2005) showed that no universal C can satisfy

$$\|(\mathcal{E}_n x)_n\|_{L_1(\mathcal{M},\ell_\infty)} \leq C \|x\|_{H_1(\mathcal{M})}$$

for all noncommutative martingales.

Hong, Junge and Parcet (2016) obtained an asymmetric maximal substitute.

Junge and Xu's question

The converse direction of the endpoint inequality is the central problem.

Problem 16, Junge and Xu (2005)

Does there exist a universal constant $C > 0$ such that

$$\|x\|_{H_1(\mathcal{M})} \leq C \, \|(\mathcal{E}_n x)_n\|_{L_1(\mathcal{M},\ell_\infty)}$$

for every noncommutative martingale?

The scalar proof uses pointwise comparisons and stopping times.

- In the classical setting, it is easy to use $M_n(f) = \sup_{1\leq k\leq n} |\mathbb{E}_k(f)|$.
- One can use the stopping time: $\nu := \inf\{n : |\mathbb{E}_k(f)| > \lambda\}$.

Our result answers this question for regular filtrations.

Davis-type inequalities

For a finite L_2-martingale, define

$$s_c(x) = \left(|d_1 x|^2 + \sum_{n \geq 2} \mathcal{E}_{n-1}(|d_n x|^2) \right)^{1/2}.$$

The conditioned and diagonal Hardy norms are

$$\|x\|_{h_p^c} = \|s_c(x)\|_p, \qquad \|x\|_{h_p^d} = \left(\sum_{n \geq 1} \|d_n x\|_p^p \right)^{1/p}.$$

Perrin (2009)

For $x \in H_1^c(\mathcal{M})$,

$$\|x\|_{H_1^c} \simeq \inf_{x=y+z} \left\{ \|s_c(y)\|_1 + \sum_{n \geq 1} \|d_n z\|_1 \right\},$$

Here y, z are martingales, and the filtration is arbitrary.

M. Perrin, J. Lond. Math. Soc. **80** (2009), 627–648.

Davis-type inequalities

For $1 \le p < 2$, let $1/p = 1/2 + 1/r$. Define the column diagonal norm

$$\|(a_n)\|_{L_p(\mathcal{M};\ell_1^c)} = \inf_{a_n = b_n c_n} \left(\sum_n \|b_n\|_2^2 \right)^{1/2} \left\| \left(\sum_n |c_n|^2 \right)^{1/2} \right\|_r,$$

and set $\|z\|_{h_p^{1c}} = \|(d_n z)_n\|_{L_p(\mathcal{M};\ell_1^c)}$.

Junge–Perrin (2014)

For $1 \le p < 2$, with equivalent norms,

$$H_p^c(\mathcal{M}) = h_p^c(\mathcal{M}) + h_p^{1c}(\mathcal{M}).$$

Moreover, $h_p^{1c}(\mathcal{M}) \subset h_p^d(\mathcal{M})$ contractively.

M. Junge and M. Perrin, Theory of $\mathcal{H}_p$-spaces for continuous filtrations in von Neumann algebras, Astérisque **362** (2014), §6.1.

Davis-type inequalities

Randrianantoanina–Wu–Xu (2019)

Let $1 \leq p < 2 \leq q < \infty$ and $x \in H_p^c(\mathcal{M}) \cap H_q^c(\mathcal{M})$. There exist martingales y, z with $x = y + z$ such that

$$\|y\|_{h_p^c} + \|z\|_{h_p^d} \leq 2^{5/2} \|x\|_{H_p^c},$$

and the same decomposition satisfies

$$\|y\|_{h_q^c} + \|z\|_{h_q^d} \leq Cq \|x\|_{H_q^c},$$

where C is an absolute constant.

N. Randrianantoanina, Lian Wu and Q. Xu, Noncommutative Davis type decompositions and applications, J. Lond. Math. Soc. **99** (2019), 97–126.

Regular filtrations

Definition

A filtration is regular if successive conditional expectations satisfy

$$\mathcal{E}_n(a) \leq c_{\mathrm{reg}}\, \mathcal{E}_{n-1}(a), \qquad a \geq 0,$$

with one constant c_{reg} independent of a and n.

Basic example is the operator-valued dyadic filtration

$$\mathcal{M}_n = L_\infty(\mathcal{D}_n) \bar{\otimes} \mathcal{N}, \qquad \mathcal{E}_n = \mathbb{E}_n \otimes \mathrm{id}_{\mathcal{N}}.$$

Here $\mathcal{D}_n$ is generated by intervals of length 2^{-n}, and

$$c_{\mathrm{reg}} = 2.$$

Main result

Theorem 1.2 (Jiao–Luo–Sukochev–Zhou)

Let $(\mathcal{M}, \tau)$ be a noncommutative probability space with a regular filtration $(\mathcal{M}_n)_{n\geq 1}$. For every martingale $x = (x_n)_{n\geq 1}$ and every $0 < p \leq 1$,

$$\boxed{\|x\|_{H_p(\mathcal{M})} \leq C_{p,\mathrm{reg}} \, \|(x_n)_n\|_{L_p(\mathcal{M},\ell_\infty)}.}$$

The constant depends only on p and the regularity constant.

At $p = 1$, this gives an affirmative answer to Junge and Xu's question under regularity.

Remarks on the main theorem

- The estimate holds throughout $0 < p \leq 1$, including the quasi-Banach range $0 < p < 1$.
- The regularity assumption is essential to the present argument. The manuscript leaves the case of arbitrary filtrations open.
- The same argument extends to semifinite von Neumann algebras equipped with a regular filtration.
- Applications include Mei's operator-valued Hardy spaces and a dyadic Carleson decomposition for noncommutative BMO.

Ingredients of the proof

The argument combines estimates for stopping projections with a column–row decomposition.

1. Cuculescu projections describe the part of a positive martingale above a given level.
2. The Fuglede–Kadison determinant and Carleman's inequality control the sum of projection tails by a positive majorant.
3. Triangular truncations split the martingale into column and row pieces. A regularity estimate controls their square function distributions.

Summing the resulting dyadic estimates yields the H_p bound.

Cuculescu projections

Let $x_n = \mathcal{E}_n(x)$ be a positive martingale. For $\lambda > 0$, set $q_0^{(\lambda)} = 1$ and

$$q_n^{(\lambda)} = q_{n-1}^{(\lambda)} \chi_{[0,\lambda]}\left(q_{n-1}^{(\lambda)} x_n q_{n-1}^{(\lambda)}\right).$$

The projections decrease in n, belong to $\mathcal{M}_n$, and satisfy

$$q_n^{(\lambda)} x_n q_n^{(\lambda)} \leq \lambda q_n^{(\lambda)}.$$

Writing $q^{(\lambda)} = \bigwedge_n q_n^{(\lambda)}$, we have

$$\lambda\tau(1 - q^{(\lambda)}) \leq \tau\left((1 - q^{(\lambda)})x\right) \leq \|x\|_1 .$$

The quantity $\tau(1 - q^{(\lambda)})$ plays the role of a maximal distribution function.

Fuglede–Kadison determinant

For a positive invertible $b \in e\mathcal{M}e$, use the restricted, unnormalized trace:

$$\Delta_e(b) = \exp\big(\tau(\log_e b)\big).$$

Lemma 3.4 (Corner estimate)

Let $e = p + q$, where p and q are orthogonal projections. If $b \in (e\mathcal{M}e)_+$ is invertible and $b \geq \lambda p$, then

$$\Delta_e(b) \geq \lambda^{\tau(p)} \Delta_q(qbq).$$

For $b = \begin{pmatrix} a & c \\ c^* & d \end{pmatrix}$, the Schur complement satisfies $s = a - cd^{-1}c^* \geq \lambda p$.
Factorization gives

$$\Delta_e(b) = \Delta_p(s)\Delta_q(d).$$

A distribution estimate from a majorant

Suppose $0 \le x_n \le a$ for every n, and define

$$A(t) = \exp\left(\frac{1}{t}\int_0^t \log \mu(s,a)\,ds\right), \qquad t > 0,$$

where $\mu(\,\cdot\,, a)$ is the generalized singular value function.

Lemma 4.4

For every $\lambda > 0$,

$$\tau(1 - q^{(\lambda)}) \le \left|\{t > 0 : A(t) \ge \lambda\}\right|.$$

For $r = 1 - q_N^{(\lambda)}$, iteration of the corner estimate gives

$$\lambda^{\tau(r)} \le \Delta_r(rar) \le \exp\left(\int_0^{\tau(r)} \log \mu(s,a)\,ds\right).$$

This implies $\lambda \le A(\tau(r))$, and A is decreasing.

Summing the projection tails

Carleman's inequality gives

$$\int_0^\infty \exp\left(\frac{1}{t}\int_0^t \log f(s)\,ds\right)\,dt \le e\int_0^\infty f(t)\,dt.$$

Applying it to $f(t) = \mu(t,a)^p$ yields

$$\int_0^\infty A(t)^p\,dt \le e\,\|a\|_p^p.$$

Lemma 4.5

For a positive majorant $x_n \le a$ and $0 < p \le 1$,

$$\sum_{k\in\mathbb{Z}} 2^{kp}\tau(1-q^{(2^k)}) \le \frac{2^p e}{2^p-1}\,\|a\|_p^p.$$

These determinant and summation estimates do not require regularity.

The regularity estimate

The additional input is a noncommutative form of Gundy's inequality.

Lemma 5.2

For the Cuculescu projections of a positive martingale associated with a regular filtration,

$$\tau\big((1-q^{(\lambda)})x\big) \le 2c_{\mathrm{reg}}\,\lambda\,\tau(1-q^{(\lambda)}).$$

This estimate replaces a trace involving the terminal value x by the size of the stopping projection.

This is where regularity enters the square function argument.

The column–row decomposition

Construct increasing projections

$$e_{j,n} = \bigwedge_{k \geq j} q_n^{(2^k)}, \qquad \pi_{0,n} = e_{0,n}, \quad \pi_{j,n} = e_{j,n} - e_{j-1,n} \ (j \geq 1).$$

For each n, the $\pi_{j,n}$ are mutually orthogonal and sum to 1.

Triangular truncations at the preceding filtration level give

$$d_n x^c = \sum_{i \leq j} \pi_{i,n-1} \, d_n x \, \pi_{j,n-1},$$

$$d_n x^r = \sum_{i > j} \pi_{i,n-1} \, d_n x \, \pi_{j,n-1}.$$

Since the projections are in $\mathcal{M}_{n-1}$, both are martingale difference sequences, and

$$x = x^c + x^r.$$

Square function distribution estimates

Write

$$B_k = \tau(1 - q^{(2^k)}), \qquad Q_\ell = \left\| q^{(2^\ell)} x q^{(2^\ell)} \right\|_2^2,$$

and $\mathrm{d}(s, b) = \tau(\chi_{(s,\infty)}(|b|))$.

Lemma 5.3

For a positive $x \in L_2(\mathcal{M})$ and a regular filtration,

$$\mathrm{d}(2^\ell, S_c(x^c)) \le C_{\mathrm{reg}} \left(2^{-2\ell} Q_\ell + \sum_{k \ge \ell} B_k \right),$$

$$\mathrm{d}(2^\ell, S_r(x^r)) \le C_{\mathrm{reg}} \left(2^{-2\ell} Q_\ell + \sum_{k \ge \ell} B_k \right).$$

The bound separates a compressed L_2 term from a projection tail.

Completion of the H_p estimate

For a positive martingale, choose a positive majorant $x_n \le a$. Proposition 4.6 gives

$$\sum_{\ell\in\mathbb{Z}} 2^{\ell(p-2)} Q_\ell + \sum_{\ell\in\mathbb{Z}} 2^{\ell p} \sum_{k\ge\ell} B_k \le C_p \|a\|_p^p .$$

The dyadic distribution formula therefore yields

$$\begin{aligned} \|S_c(x^c)\|_p^p &\lesssim_p \sum_{\ell\in\mathbb{Z}} 2^{\ell p} \mathrm{d}(2^\ell, S_c(x^c)) \\ &\le C_{p,\mathrm{reg}} \|a\|_p^p . \end{aligned}$$

The same bound holds for $S_r(x^r)$. Taking the infimum over majorants,

$$\|x\|_{H_p} \le C_{p,\mathrm{reg}} \|(x_n)_n\|_{L_p(\mathcal{M},\ell_\infty)} .$$

Positive decomposition gives the extension to general martingales.

Operator-valued Hardy spaces

Let $(\mathcal{N}, \varphi)$ be semifinite and set

$$\mathcal{M} = L_\infty(\mathbb{R}) \bar{\otimes} \mathcal{N}.$$

For the Poisson extension $P_\varepsilon f$, define

$$G_c(f)(s) = \left(\int_0^\infty \varepsilon \left| \partial_\varepsilon P_\varepsilon f(s) \right|^2 \, d\varepsilon \right)^{1/2}.$$

Mei's spaces satisfy

$$\|f\|_{\mathcal{H}_1^c} = \|G_c(f)\|_1, \qquad \mathcal{H}_1 = \mathcal{H}_1^c + \mathcal{H}_1^r, \qquad \|f\|_{\mathcal{H}_1^r} = \|f^*\|_{\mathcal{H}_1^c}.$$

For $h_1, h_2 > 0$, consider the interval averages

$$A_h f(t) = \frac{1}{h_1 + h_2} \int_{t-h_1}^{t+h_2} f(s) \, ds.$$

The operator-valued Hardy estimate

Theorem 6.3 (Operator-valued Hardy estimate)

For every $f \in L_1(\mathcal{M})$,

$$\|f\|_{\mathcal{H}_1(\mathbb{R},\mathcal{N})} \leq C \left\| (A_h f)_{h \in \mathbb{R}_+^2} \right\|_{L_1(\mathcal{M},\ell_\infty)},$$

where C is universal.

Use the dyadic filtration, whose regularity constant is 2:

$$\begin{aligned} \|f\|_{\mathcal{H}_1(\mathbb{R},\mathcal{N})} &\lesssim \|f\|_{H_1(\mathcal{M},\mathcal{D})} \\ &\lesssim \|(\mathbb{E}_n f)_n\|_{L_1(\mathcal{M},\ell_\infty)} \\ &\lesssim \|(A_h f)_h\|_{L_1(\mathcal{M},\ell_\infty)}. \end{aligned}$$

The first step is Mei's Hardy space comparison. The second step is our Davis inequality.

Noncommutative martingale BMO

For a noncommutative probability space, define

$$\|f\|_{\mathcal{BMO}^c} = \sup_{n\geq 1} \left\| \mathcal{E}_n\left(|f-\mathcal{E}_{n-1}f|^2\right)\right\|_\infty^{1/2}, \qquad \mathcal{E}_0 f = 0,$$

and

$$\|f\|_{\mathcal{BMO}^r} = \|f^*\|_{\mathcal{BMO}^c}.$$

The two-sided space is

$$\mathcal{BMO}(\mathcal{M}) = \mathcal{BMO}^c(\mathcal{M}) \cap \mathcal{BMO}^r(\mathcal{M}),$$

with

$$\|f\|_{\mathcal{BMO}} = \max\{\|f\|_{\mathcal{BMO}^c}, \|f\|_{\mathcal{BMO}^r}\}.$$

The duality $H_1(\mathcal{M})^* = \mathcal{BMO}(\mathcal{M})$ connects the Davis inequality with positive BMO decompositions.

A positive BMO decomposition

Theorem 6.6

Assume that the filtration is regular. For every $f \in \mathcal{BMO}(\mathcal{M})_+$, there exist $g \in \mathcal{M}_+$ and $u_k \in \mathcal{M}_+$ such that

$$f = g + \sum_{k \geq 1} \mathcal{E}_k(u_k) \qquad \text{in the strong operator topology,}$$

and

$$\|g\|_\infty + \left\| \sum_{k \geq 1} u_k \right\|_\infty \leq C_{\mathrm{reg}} \|f\|_{\mathcal{BMO}} .$$

Talebi, Moslehian and Sadeghi (2019) obtained this representation assuming the endpoint Hardy–maximal estimate.

Our theorem supplies that estimate for regular filtrations.

Dyadic Carleson decomposition

Let $\mathcal{M} = L_\infty(0,1)\bar{\otimes}\mathcal{N}$ with the dyadic filtration, where $(\mathcal{N}, \varphi)$ is a noncommutative probability space.

Theorem 6.7

For every $f \in \mathcal{BMO}(\mathcal{M})_+$, there exist $g \in \mathcal{M}_+$, dyadic intervals I_i, and $a_i \in \mathcal{N}_+$ such that

$$f = g + \sum_i a_i \chi_{I_i}, \qquad \|g\|_\infty \le \|f\|_{\mathcal{BMO}},$$

and, for every dyadic interval I_0,

$$\left\| \frac{1}{|I_0|} \sum_{i:\, I_i \subset I_0} a_i |I_i| \right\|_{\mathcal{N}} \le C \|f\|_{\mathcal{BMO}}.$$

Here C is universal.

This is the operator-valued counterpart of the dyadic Garnett–Jones decomposition.

Selected references

- B. Davis (1970). *On the integrability of the martingale square function*. Israel J. Math. **8**, 187–190.
- M. Junge and Q. Xu (2005). *On the best constants in some non-commutative martingale inequalities*. Bull. London Math. Soc. **37**, 243–253.
- Y. Jiao, N. Randrianantoanina, L. Wu and D. Zhou (2020). *Square functions for noncommutative differentially subordinate martingales*. Comm. Math. Phys. **374**, 975–1019.
- T. Mei (2007). *Operator valued Hardy spaces*. Mem. Amer. Math. Soc. **188**, no. 881.
- A. Talebi, M. S. Moslehian and G. Sadeghi (2019). *A representation of noncommutative BMO spaces*. Statist. Probab. Lett. **153**, 65–70.
- J. B. Garnett and P. W. Jones (1982). *BMO from dyadic BMO*. Pacific J. Math. **99**, 351–371.

Thank you!